Understanding the Time–Space Tradeoff in Traffic Forecasting: Evidence from Sequential and Graph Neural Networks

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Keywords: Traffic Detection, Machine Learning, Tabular Gradient-Boosted Models, XGBoost, LightGBM, Deep Sequential Models, GRU, TCN, LSTM, Spatiotemporal Graph Neural Networks, LSTM-SAGE, GNN, Spatio-Temporal Models, F1-score.

Abstract

This study develops a real-time early-warning system for predicting sudden traffic breakdowns across the San Francisco Bay Area highway network using the PeMS-BAY dataset, comprising approximately 16 million speed readings from 325 loop-detector sensors collected over six months. A broad spectrum of machine learning architectures was systematically evaluated, ranging from tabular gradient-boosted models (XGBoost, LightGBM) to deep sequential models (GRU, TCN, LSTM) and Spatio-Temporal Graph Neural Networks (LSTM-SAGE GNN), across both general steady-state traffic and isolated congestion transition events. XGBoost achieved a dominant F1-score of 0.8913 on continuous flow prediction, confirming that traffic inertia makes historical speed a strong general predictor. However, when focusing on the more compact dataset focusing on sudden changes in traffic, spatio-temporal models proved superior, achieving a leading F1-score of 0.7112. While other models such as the purely temporal modes such as the TCN and GRU outperformed GNNs in lower epoch runs, the added spatial complexity ended up giving GNNs an edge. This suggests that for more complex traffic data GNNs are superior due to their ability to manage spatial complexity. Lastly further analysis on GNN radii with the regular dataset showed peak performance at around the 2 mile radius with an F1 score of .7304 displaying how smaller radii GNNs are better able to separate noise from important data and should be considered when mapping out traffic.

1. Introduction

Road infrastructure forms the backbone of the United States economy, with approximately 72% of all domestic freight transported by trucks traveling the interstate highway system. Yet despite this critical role, rapidly growing travel demand and finite physical capacity have left the nation's road network severely strained. With this issue producing cascading congestion delays in commercial shipping and everyday commuter travel alike. Expanding physical capacity through new construction is a highly resource intensive and often impractical undertaking. As such, an alternative solution lies in maximizing the efficiency of existing infrastructure. The volume of data now generated by traffic monitoring systems, autonomous vehicles, and embedded road sensors has created an unprecedented opportunity to manage this demand. Machine learning offers a path forward through the creation of data-driven systems capable of predicting and mitigating severe congestion bottlenecks before they occur.

Despite the volume of traffic sensor data now available through systems such as the California Performance Measurement System (PeMS-BAY), there remains a limited methodological understanding of which machine learning architectures are best suited for specific traffic forecasting tasks. Existing literature frequently evaluates models based on continuous speed prediction. However in this regression task simple baselines often perform deceptively well due to high traffic inertia and as such largely neglect the distinct challenge of predicting the precise chronological onset of a localized breakdown. Consequently, it remains unclear whether computationally heavy spatio-temporal graph architectures actually provide a tangible advantage over highly optimized tabular or deep sequential models when specifically targeting these volatile transition states. This ambiguity limits the actionability of current research for real-world, early-warning deployment.

This paper aims to address this methodological gap by benchmarking a diverse suite of machine learning models ranging from tabular gradient boosting to deep sequential networks and spatio-temporal graph architectures on the PeMS-BAY highway sensor dataset. To rigorously evaluate architectural suitability, the models are tested on both the standard dataset and a concentrated subset focused exclusively on complex transition moments. Formally, this study reframes traffic forecasting from a continuous regression task to a discrete binary classification problem. Using short term historical time series speed data augmented by the geographic topological graph structure of the sensor network, the models are tasked with mapping these inputs to a binary output. The objective is to classify the future state of a specific road segment 15 minutes into the future as either congested (below 60 miles per hour traffic) or uncongested, thereby isolating and comparing the models' specific capabilities to anticipate critical bottlenecks before they fully materialize.

2. Literature Review

Road infrastructure management faces three compounding challenges that make AI-driven solutions both necessary and timely. First, aging roads and deferred maintenance lead to progressive degradation, producing potholes, surface wear, and safety hazards that compound costs over time. Second, inefficient resource planning and poor data utilization can result in cost overruns and suboptimal project execution. Third, the absence of real-time, scalable monitoring systems makes early detection of damage difficult, forcing agencies into expensive reactive repairs rather than preventive maintenance. These structural inefficiencies have made data-driven, machine learning-based approaches an active area of research across infrastructure engineering and intelligent transportation systems.

Early attempts at automated road damage detection relied on traditional image processing techniques edge detection, thresholding, and texture analysis but these methods were brittle against variations in lighting conditions, weather, and pavement type. The introduction of Convolutional Neural Networks (CNNs) marked a foundational breakthrough [1], enabling automatic feature extraction directly from raw images without hand-engineered descriptors. A comprehensive evaluation of eight CNN-based damage detection models including SSD MobileNet [2, 3], Faster R-CNN Inception [4], and Faster R-CNN ResNet variants found that Faster R-CNN Inception V2 provided the strongest balance of accuracy and real-time processing speed across five road damage categories including longitudinal cracks, potholes, and alligator damage [5]. This work established CNN-based detection as the dominant paradigm for automated road surface assessment.

Building on this foundation, Ferrari et al. [6] extended CNN-based analysis to high-resolution 3D street-level imagery for large-scale municipal infrastructure assessment, while Ejaz et al. [7] developed a hybridized edge-cloud architecture that computes road damage locally on embedded devices before aggregating results to the cloud, reducing latency while preserving detection accuracy. More recently, an integrated YOLO [8]-CNN-BiLSTM [9] framework combining multimodal IoT sensing with a Digital Twin simulation layer demonstrated precision of 96.26% for real-time pothole and crack detection, outperforming prior state-of-the-art models while also predicting deterioration trends not merely detecting current damage states [10].

Despite their strong spatial feature extraction capabilities, standard CNNs are inherently limited to frame-level analysis and cannot capture how road conditions evolve over time. To address this, researchers began integrating CNNs with Long Short-Term Memory networks (LSTMs) [11], which are designed to model sequential and

time-dependent patterns. Lyu et al. [12] explored CNN-LSTM architectures for road segmentation tasks, demonstrating that combining spatial feature extraction with temporal sequence modeling improved performance on continuous video-based road monitoring. Extending this direction, Hadj-Attou et al. proposed two hybrid architectures, a CNN-GRU [13] and a CNN-LSTM [11] for classifying road surface anomalies from vehicle sensor streams, with both models outperforming their non-temporal CNN counterparts [14]. These hybrid systems have been particularly effective in vehicle-mounted monitoring applications, where cameras and accelerometers continuously collect data and the model must recognize how damage progresses between frames. A critical limitation of this architecture family, however, is that it models each road segment in isolation and as such damage detected at one point in a network is not propagated or contextualized against conditions at neighboring segments.

To overcome this relational blindspot, researchers began modeling road infrastructure as graphs, where intersections or sensors serve as nodes and road segments or adjacency relationships serve as edges. Graph Neural Networks (GNNs) [15] are naturally suited to this representation, as they propagate information across connected nodes and can model how deterioration or congestion in one segment influences its neighbors. Hamilton et al. [16] introduced GraphSAGE (Graph Sample and Aggregate), which enabled scalable inductive representation learning on large graphs. Liu and Liu [17] extended this perspective with a systematic review confirming that GNNs improve both reliability assessment and resilience modeling of interconnected infrastructure systems compared to non-graph architectures.

In the traffic forecasting domain, Luo et al.'s comprehensive STG4Traffic benchmark [18] which evaluated five mainstream spatio-temporal GNN models including DCRNN [19], Graph WaveNet [20], and T-GCN [21] on the PeMS-BAY dataset family confirmed that GNNs consistently achieve state-of-the-art mean absolute error on general speed prediction tasks by exploiting the high spatial correlation between neighboring sensors. Forkan et al. [22] demonstrated a complementary direction, using AI and IoT-driven city-scale sensing to monitor roadside infrastructure conditions in real time, further establishing that graph-structured spatial modeling is well-suited to large-scale network assessment.

The most recent wave of research has focused on spatio-temporal GNNs (ST-GNNs), which integrate both spatial graph convolution and temporal sequence modeling effectively combining the strengths of CNNs, LSTMs, and graph-based learning in a single architecture. Jin et al. [23] proposed the Spatio-Temporal Graph Neural Point Process (STGNPP) specifically for traffic congestion event prediction, framing congestion onset as a point process on a graph rather than a continuous regression task. Their results highlighted that while ST-GNNs accurately model broad

congestion propagation waves across a network, they struggle to predict the precise chronological moment a localized jam begins, generating elevated false positives at the sensor level.

This limitation is echoed in research on temporal convolutional approaches to traffic anomaly detection. Ren et al. [24] proposed a graph autoencoder combining a Mirror Temporal Convolutional Network (TCN) [25] with a Graph Convolutional GRU [13] to detect anomalies in traffic time series, finding that the dilated convolution structure of TCNs which can attend to long-range temporal dependencies in parallel outperformed purely spatial architectures on sparse, localized anomaly events. A direct comparison of deep learning and ensemble approaches by Zheng et al. [26] further found that XGBoost [27], using only temporal features engineered from historical traffic readings, performs competitively with Transformer-based deep learning models [28] on long-horizon traffic forecasting a result with direct implications for infrastructure monitoring tasks where labeled breakdown events are rare.

Collectively, the literature reveals a clear and underexplored gap: while ST-GNNs dominate general traffic speed prediction benchmarks, their superiority does not extend uniformly to the specific task of predicting the exact onset of localized congestion transitions. Furthermore, existing GNN research on road infrastructure has concentrated heavily on highway scale macro-networks, leaving the question of optimal spatial connection radius and its effect on breakdown prediction accuracy largely unaddressed.

3. Methodology

Dataset Description

The primary dataset used to research traffic congestion primarily in the bay area was the PeMS-BAY(Performance Measurement System/Freeway Performance Measurement System) dataset [29], a dataset containing the speed of cars over 325 unique loop-detector sensors. These loop-detectors were spread across major highways in the bay area transportation system and used magnetic sensors embedded inside of the road in order to calculate the average speed of cars going over these sensors. The dataset consisted of the average traffic speeds at all of these detectors every 5 minutes over a 6 month period from January 2017 to June 2017. Over this period of time sensors gathered approximately 16 to 17 million individual data points that were used to gather data. In addition to this dataset, the geographic location of all of the sensors was also taken for further analysis as well as incorporated into other models. Figure 1 shows the geographic locations of the sensors in the PeMS-BAY dataset surrounded by satellite imagery.

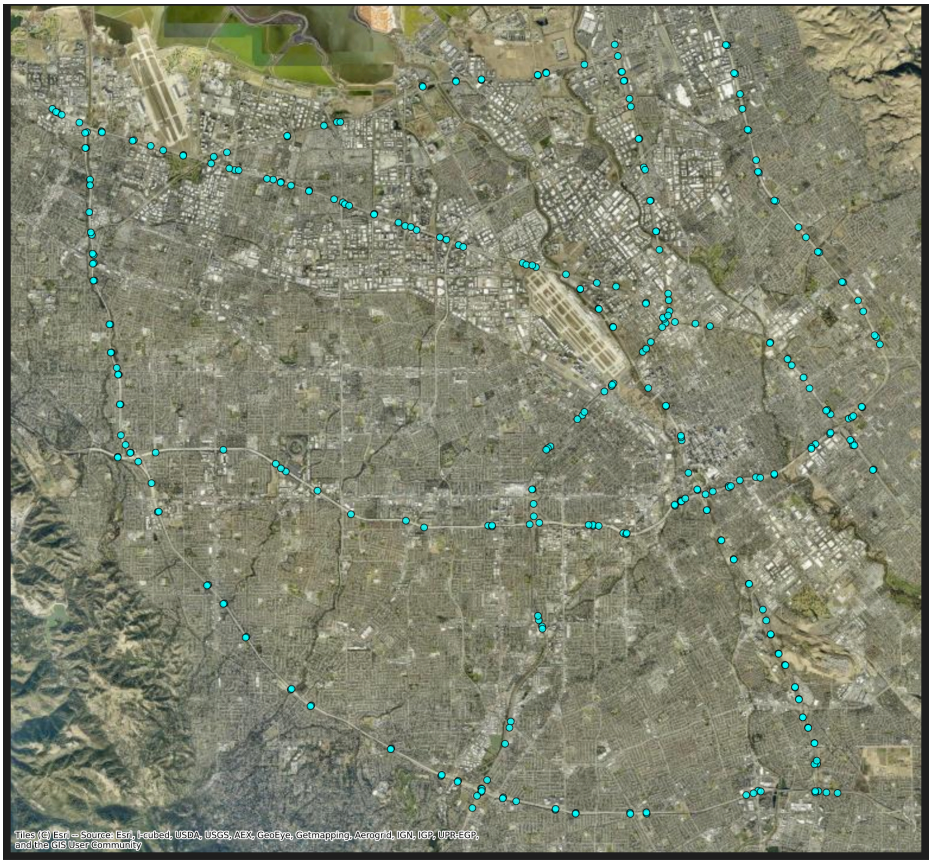


Figure 1: Overview of the 325 sensor data in the bay area shown in satellites. Satellites show these sensor locations gathering data.

Out of the roughly 16 million datapoints there were roughly 500 times in which the data sensors showed 0 mph as the current speed. Upon further analysis these data points were shown to have been caused by mass outages at certain times during the day. As a result these datapoints were replaced by a rolling mean when cleaning up the dataset. Furthermore, initial models were often heavily focused on computing more trivial data where traffic congestion does not change sharply. As such a second dataset was developed from the PeMS-BAY dataset that focused more heavily on when each node shifted from congested (less than 60mph) to uncongested or vice versa.

Methods

Since a large majority of interstate systems inside of the bay area are composed of freeways. The intended average speed over the PeMS-BAY network is thus 70 mph and as such highway networks where cars are moving at around an average of 60 miles per hour can be considered slightly congested. As such models were tested on the

regular PeMS-BAY dataset and the concentrated dataset focusing on changes in speed from those above and below 60 miles per hour. The models were thus evaluated on not only how well they performed on regular traffic data but also in detecting shifts in traffic.

A wide variety of models were tested including simpler ones such as Logistic Regression and Random Forest algorithms. From there tabular models such as XGBoost and LightGBM were tested next. These tabular models create decision trees to predict outcomes. Regular Graph Neural Networks(GraphSAGE and GAT) which relied on geographic location to determine nodes and edges were tested due to the physical features of traffic propagating throughout the network. Temporal models such as the LSTM,GRU and TCN ,able to store time based information using memory, were tested to check for the time-based side of traffic prediction. Spatio-temporal models,the leading model for traffic prediction and analysis, combined these two features and was tested as well. For GNNs, temporal models, and spatio-temporal models PyTorch and PyTorch Geometric were used in order to build custom models.

To evaluate the models, the dataset was partitioned chronologically to prevent data leakage. The tabular baseline and sequential deep learning models (GRU, LSTM, TCN, Transformer) utilized a 70/15/15 split for training, validation, and testing, respectively. The TCN was designed with four sequential residual blocks utilizing a fixed kernel size of 5 and exponentially increasing dilation factors (d = 1, 2, 4). This architecture yields a maximum theoretical receptive field of 61 time steps, ensuring the network comprehensively captures the entire 12-step historical input sequence before making a prediction. Due to the computational demands of the graph structures, the spatio-temporal LSTM-GNN models utilized an 80/20 training-validation split. All deep learning models were optimized using the Adam optimizer and trained using a binary cross-entropy loss function. The sequential models (GRU and LSTM: 2 layers, 64 hidden units) were initialized with a learning rate of 0.0005 and utilized a dynamic learning rate scheduler (ReduceLROnPlateau) that halved the learning rate if the validation metric stagnated for 15 epochs. To prevent overfitting while allowing sufficient time for convergence, the sequential models were permitted to train for up to 500 epochs, governed by an early stopping mechanism that halted training if the validation F1 score failed to improve for 50 consecutive epochs. Conversely, the spatio-temporal LSTM-GNN models sequentially projected a 16-dimensional temporal LSTM embedding through two 32-dimensional GraphSAGE convolutional layers. These models utilized a higher initial learning rate of 0.005 and a stricter early stopping patience of 10 epochs over a maximum of 200 epochs due to higher usage of resources to train one epoch. For the tabular baselines (e.g. Gradient Boosting), models were constrained to a maximum of 200 iterations with a learning rate of 0.05, utilizing a 15% internal validation fraction to govern early stopping.

To evaluate the efficacy of the early-warning traffic forecasting models, we relied primarily on the F1-score, Precision, and Recall, while supplementing with overall Accuracy and ROC-AUC. Traffic breakdown events are inherently rare anomalies within the broader continuous dataset. Thus relying solely on accuracy is highly misleading in this context as a naive baseline model predicting that traffic will simply maintain its current state achieves ~85% accuracy due to the physical inertia of traffic. Therefore, the F1-score was selected as our primary optimization metric because it strictly balances Precision (minimizing false alarms that could erode user trust) and Recall (ensuring true impending jams are not missed). ROC-AUC was utilized during model tuning to measure the models' ability to probabilistically distinguish between steady-state flow and critical breakdown thresholds across various decision boundaries.

4. Results and Discussion

The experimental results are separated into two distinct problem domains: General Steady-State Prediction (continuous flow) and Isolated Breakdown Prediction (detecting exact transition moments).

When evaluating the models on the continuous general dataset, Tabular models dominated, with XGBoost and LightGBM being the most successful, getting an F1 score close to 0.9. However, when evaluated strictly on the concentrated dataset of isolated breakdown events, spatial temporal models and temporal models outperformed the tabular models mentioned above. Figure 2 outlines the comparison of different AI models trained to benchmark performance on the PeMS-BAY dataset.

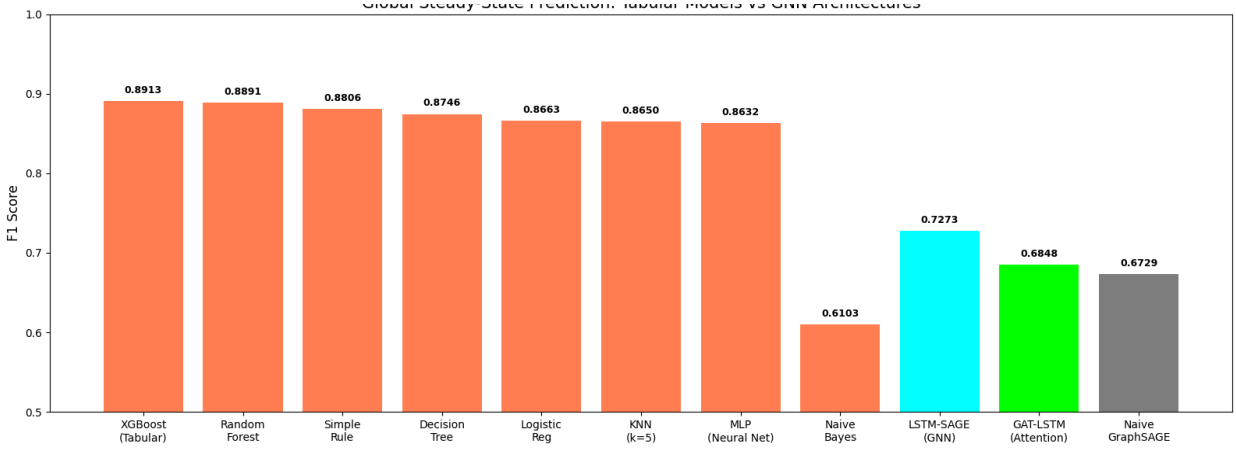


Figure 2: Global Steady-State Prediction: Tabular Models vs GNN Architectures on the larger PeMS-BAY dataset.

In Figure 3, it can be seen that for the dataset concentrated on changes in traffic flow from congested to uncongested or vice versa spatial-temporal models were able to outperform tabular models such as XGBoost and LightGBM. This development thus shows that while models such as XGBoost are able to accurately predict traffic as a whole, the temporal models are better able to predict key moments when traffic swaps from congested to uncongested or vice versa. This highlights the importance of time when attempting to find out when exactly traffic begins to shift. However originally GNNs struggled with lower amounts of epochs and as such were not on the initial graph.

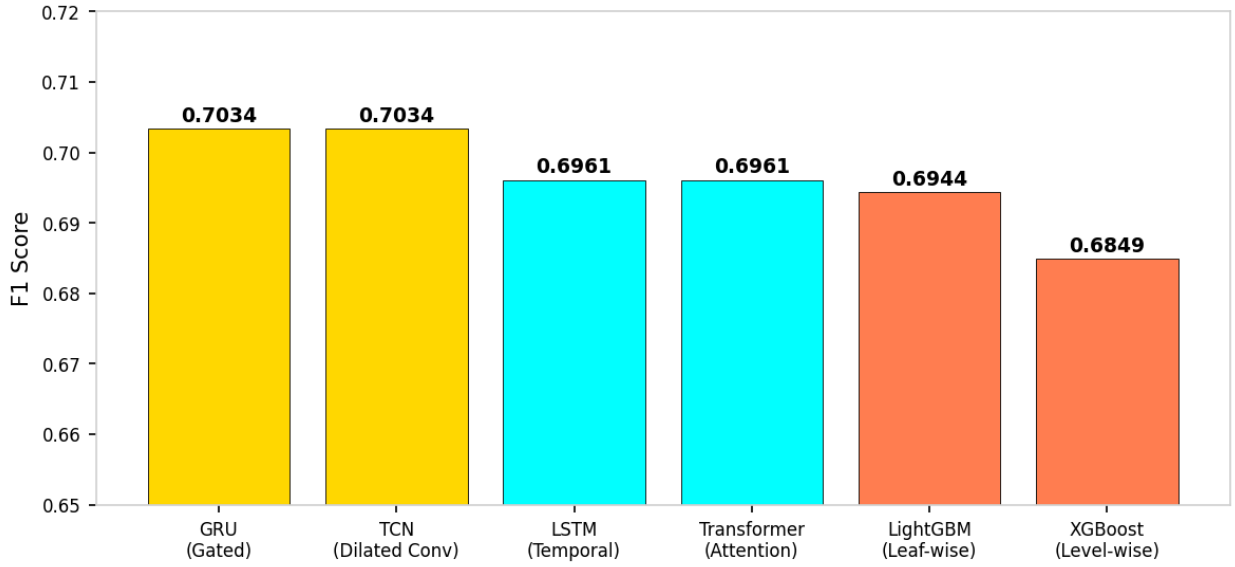


Figure 3: Transition Event Prediction comparison of models.

When trained on an extended amount of epochs, the vast majority of the models flatlined around the .6935 F1 score range, increasing their F1 scores only slightly compared to the GNNs. This suggests that the increased complexity in determining when exactly traffic stops allows GNNs to use the increased data from surrounding nodes to outcompete other models after receiving increased data. More notably the other models stopped close to each other potentially showing that certain increases in traffic are unpredictable without further information, as both the tabular models and regular temporal models stopped roughly around the same F1 score. As such even though Spatio-Temporal models struggled with modeling traffic congestion as a whole, with extended training they were able to outperform other models in detecting when exactly congestion occurred.

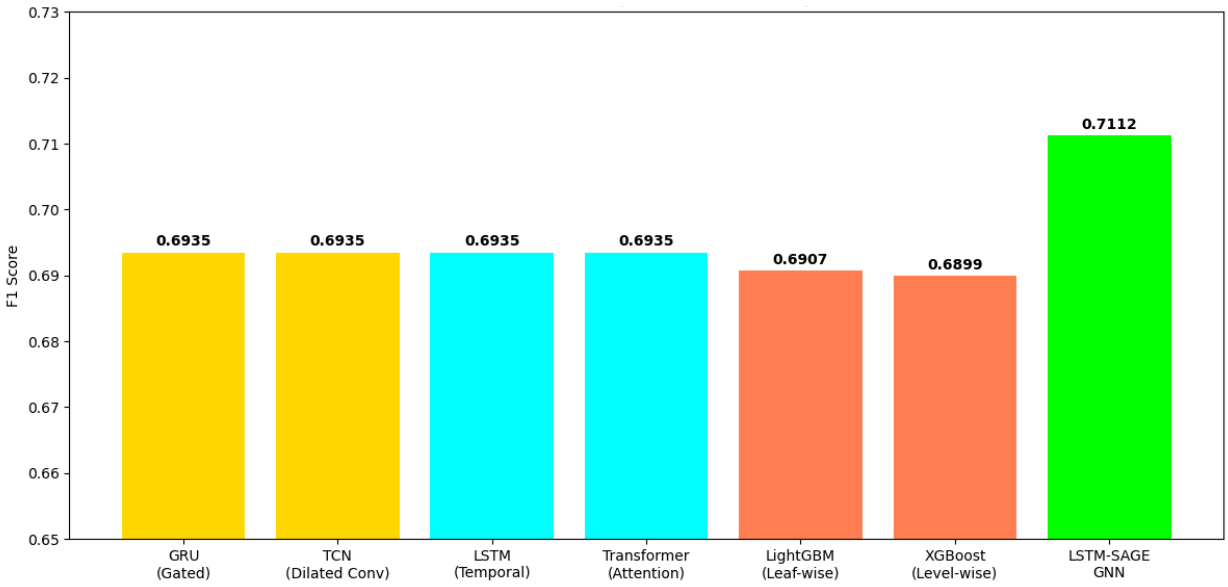


Figure 4 Transition Event Prediction models trained for extended amounts of time.

To map the spatial physics of the highway network, a Spatio-Temporal Graph Neural Network (LSTM-SAGE GNN) was tested across 12 distinct geographic connection radii ranging from 1 to 20 miles. The resulting performance curve revealed two distinct spatial correlation peaks. As shown in Figure 5, a local peak emerged at 4.5 miles (F1 = 0.7055 on continuous data), possibly due to the model being able to pick up on smaller tighter knit highway corridors and their connections. A global peak emerged at 8.0 miles (F1 = 0.7169), which was likely able to capture broader inter exchange shockwave propagation across the Bay Area network. Beyond 10 miles, the graph became overly dense, and cross-corridor noise caused performance to collapse. As such when training with fewer epochs, GNNs of larger radii were likely able to use the data gained from new nodes to create better results.

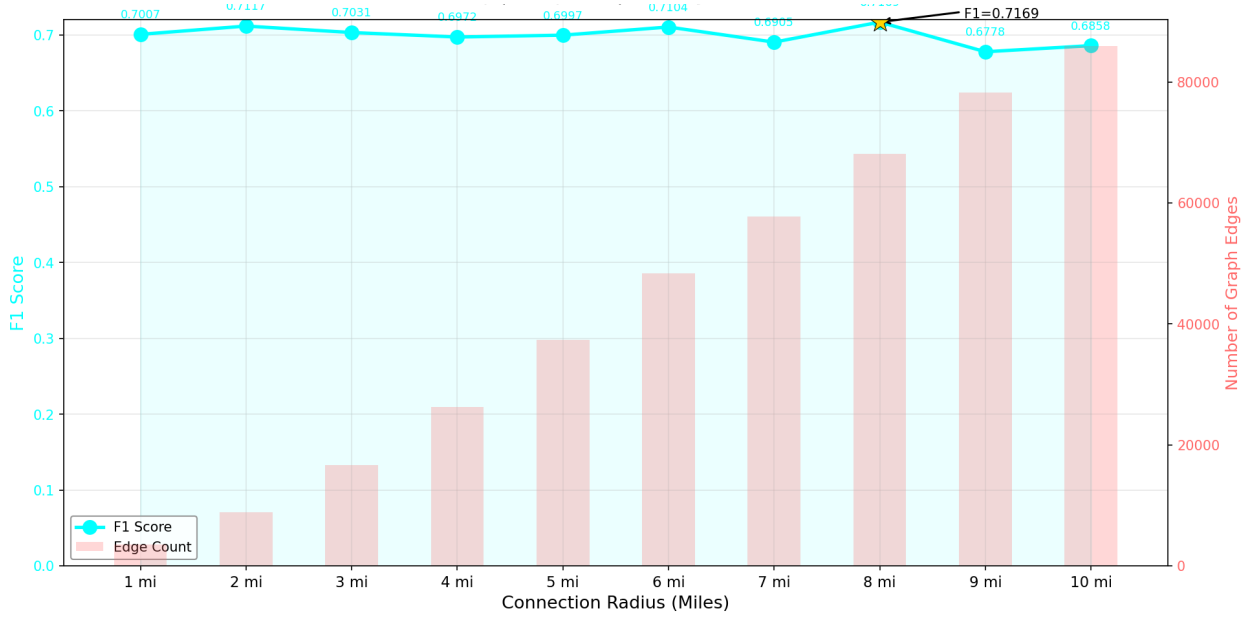


Figure 5: Spatio-Temporal GNN(LSTM-GNN) Performance with various connection radii 25 Epochs using the PeMS-BAY dataset

Figure 6 showcases the model performance over extended training epochs. The models were trained with early stopping so if the model performance starts to plateau the training cycle will stop to preserve resources and time. As the graph shows, the most substantial changes occurred within the first 10 epochs with the model plateauing quickly and only making small gains over time. This could potentially be due to the increased amount of noise that occurred in this graph. Looking at Figure 6 and 7 however we can see that overall smaller radii seemed to benefit from longer training more. Several reasons may have caused the graphs to appear this way. Firstly a wider

set of connected nodes could have amplified noise between nodes.

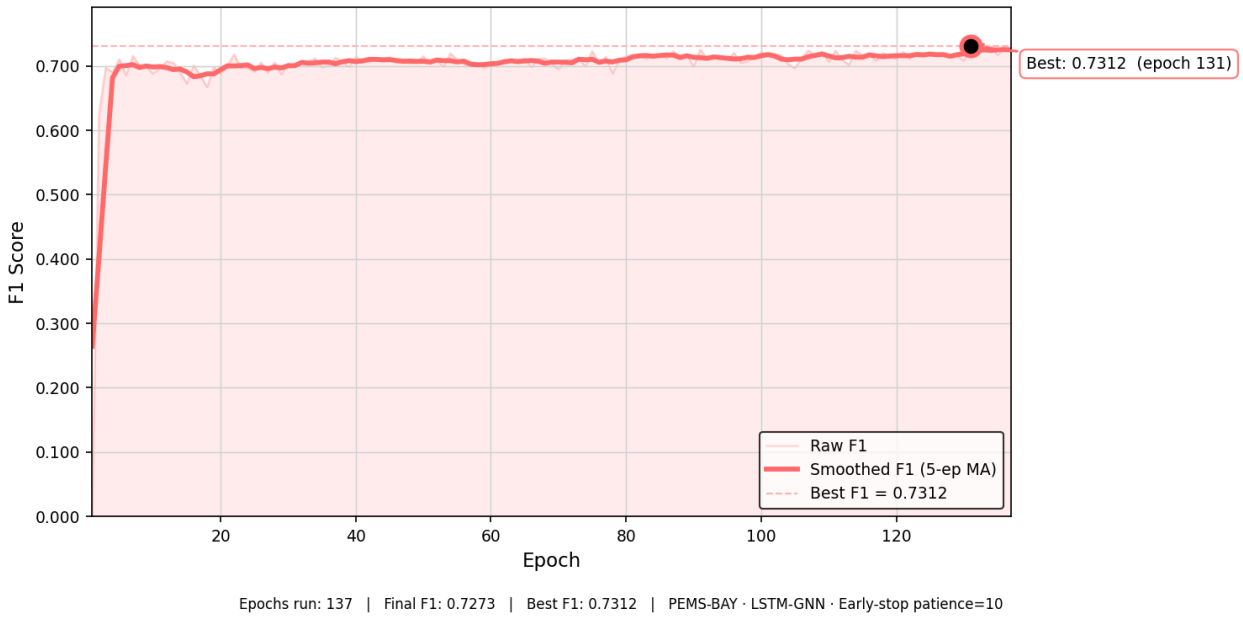


Figure 6: Model Training for extended epochs using the 2 mile connection radius

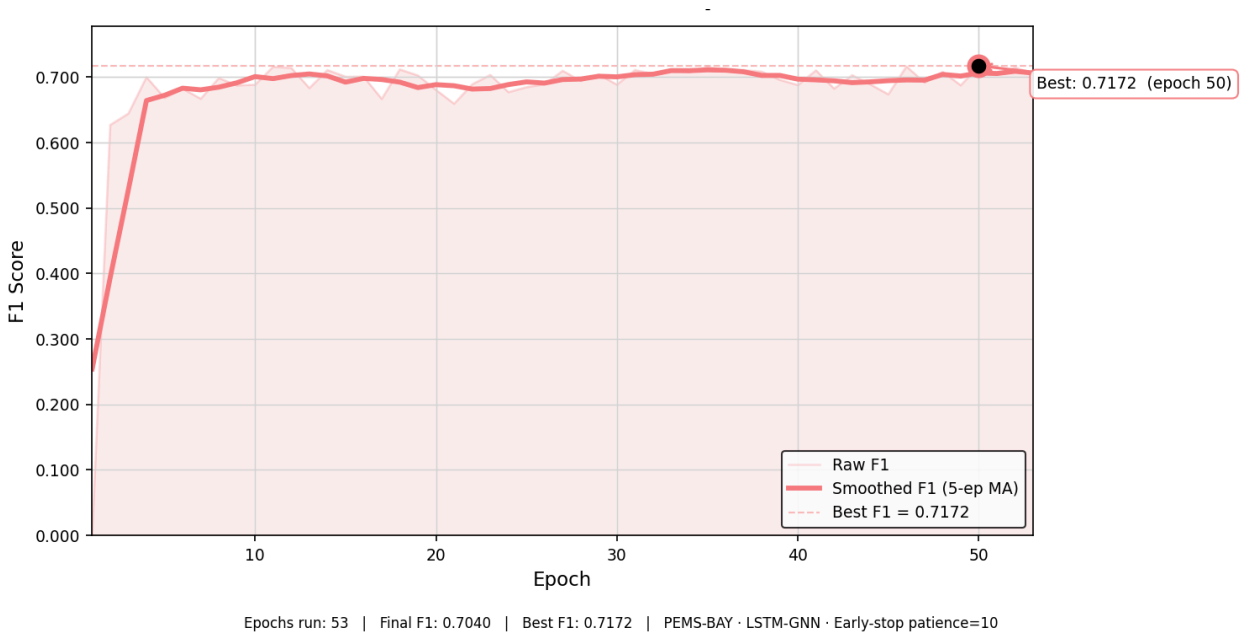


Figure 7: Model Training for extended epochs using the 10 mile radius

Figure 8 shows different radii locations along with the performance scores of the models. It also shows the sensors which are not connected during the training of a certain radius. These radii were local peaks in the initial comparisons between

spatio-temporal models. While the 2 mile radii (8.2) and 4.5(8.3) mile radii appear to rely more on closer knit sections of the highway system to determine congestion, the 10(8.4) mile radii model is covers most of the system near the center and thus is able to use more information as a whole to determine congestion in the system.



Figure 8.1: Graph of a center node with a circle of radius 1. The green dots show what nodes are connected or close to the center node.



Figure 8.2: Graph of a center node with a circle of radius 2. The blue dots show what nodes are connected or close to the center node.

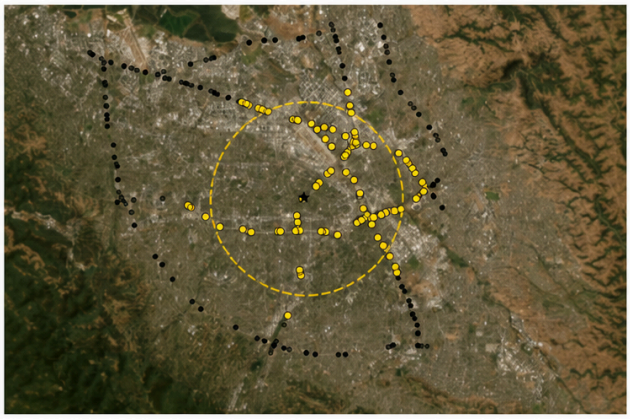


Figure 8.3: Graph of a center node with a circle of radius 4.5. The yellow dots show what nodes are connected or close to the center node.



Figure 8.4: Graph of a center node with a circle of radius 10. The red dots show what nodes are connected close to the center node.

Figure 8: Sensor connection radius visualization of various radii used for spatio-temporal models.(Split into 2 images with 2 maps each to make a 2x2 grid) Add subcategories for figures 8a,8b,8c,8d. Remove text from the graphs to increase readability.

Comparison with Past Work

Our findings regarding the dominance of tree-based models over complex deep learning architectures strongly align with the recent observations by Zheng et al. [26]. They noted that when temporal dynamics are effectively captured through rigorous feature engineering such as rolling means and temporal lags, models like XGBoost

frequently outperform deep sequence models on tabular representations of time-series data. In our study, XGBoost not only provided competitive predictive accuracy but did so with significantly lower computational overhead and faster convergence. reinforcing the viability of gradient boosting for real-time traffic forecasting.

When evaluating our spatio-temporal deep learning approach against the STG4Traffic benchmark established by Luo et al. [18], our LSTM-GNN architecture demonstrates the baseline capabilities of spatial aggregation but falls short of state-of-the-art results on large-scale, highly dynamic networks. Specifically, our peak GNN performance of 0.7179 is notably lower than the accuracies reported in recent literature such as Jin et al. [23]. This discrepancy is likely rooted in three key differences in experimental design. Firstly, while much of the literature focuses on continuous speed or flow prediction , our task is formulated as a strict binary congestion classification during highly volatile transition windows, inherently lowering the performance ceiling. Secondly, our dataset specifically isolated difficult, non-stationary transition events rather than evaluating performance over a random sampling of predictable, steady-state traffic. Finally, Model Architecture: Jin et al.[13] and the leading models in the STG4Traffic benchmark utilize highly advanced spatio-temporal attention mechanisms such as ASTGCN and Graph WaveNet that can dynamically weight edge importance. By contrast, our implementation relied on standard GraphSAGE convolution, which weights all neighbors equally and is therefore more susceptible to the spatial noise observed at larger radii.

Discussion/Interpretation

The results demonstrate that the effectiveness of a machine learning architecture depends strongly on the specific traffic prediction task being addressed. For continuous steady-state traffic prediction, gradient-boosted tabular models such as XGBoost and LightGBM substantially outperformed the deep learning architectures. This result reflects the strong temporal inertia of traffic flow, where current traffic conditions are highly predictive of conditions in the near future. In these situations, the additional complexity of sequential or graph-based models provides relatively little benefit over well-engineered tabular approaches.

The transition-event dataset presented a considerably more challenging prediction problem. Unlike continuous traffic forecasting, predicting the precise onset of congestion requires distinguishing subtle changes that precede a traffic breakdown. While temporal models such as the GRU and TCN converged quickly and achieved strong early performance, the LSTM-SAGE Graph Neural Network ultimately achieved the highest F1-score after extended training. This improvement suggests that

incorporating spatial relationships between neighboring highway sensors becomes increasingly valuable when sufficient training allows the model to learn how congestion propagates through the transportation network.

The spatial radius experiments further demonstrate the importance of graph construction when designing spatio-temporal models. Models using smaller connection radii generally benefited more from extended training, likely because they focused on highly relevant neighboring sensors while reducing the noise introduced by distant roadway segments. Although intermediate and larger radii captured broader network interactions, excessive graph connectivity reduced performance by introducing unnecessary spatial information. These findings indicate that careful graph design is as important as model architecture when applying Graph Neural Networks to traffic forecasting. For more information look at Appendix A.

Overall, the findings suggest that intelligent transportation systems should employ multiple complementary models rather than relying on a single architecture. Gradient-boosted models are highly effective for continuous traffic monitoring, while spatio-temporal Graph Neural Networks provide superior performance for identifying localized congestion transitions where spatial relationships between highway segments play a critical role.

5. Conclusion

This research developed and evaluated a real-time machine learning framework for predicting traffic conditions across the San Francisco Bay Area highway network using the PeMS-BAY dataset. Using approximately sixteen million traffic speed observations collected from 325 roadway sensors, two complementary prediction tasks were investigated: continuous steady-state traffic forecasting and the prediction of isolated congestion transition events. This approach allowed the strengths of different machine learning architectures to be evaluated under both routine and more challenging traffic conditions.

The results indicate that model effectiveness is task-dependent. XGBoost achieved the highest performance for general traffic prediction (F1 = 0.8913), whereas the LSTM-SAGE Graph Neural Network performed best for congestion onset prediction (F1 = 0.7112), highlighting the importance of incorporating both temporal and spatial dependencies when modeling traffic transitions. Furthermore, smaller graph connection radii consistently improved performance by emphasizing local traffic interactions and reducing noise from distant roadway segments. It should also be noted that GNNs took longer to train effectively as shown by GNNs falling behind in the 25 epoch runs on the

concentrated dataset but were able to vastly outperform the other models when all models were trained for 200 epochs .

Although the proposed framework achieved strong predictive performance, several limitations remain. The models were trained using only historical traffic speed data and therefore cannot directly account for external influences such as accidents, weather, construction, or special events. Furthermore, the computational cost of training large spatio-temporal graph networks limits rapid hyperparameter optimization and experimentation.

Future iterations of this system could integrate deeper transformer-based attention architectures trained on substantially larger, multi-year datasets encompassing diverse weather conditions. Further optimization of the GNN hyperparameters and dynamic edge-weighting could help mitigate spatial noise.

Deploying this predictive system in a real-world Intelligent Transportation System (ITS) requires balancing predictive accuracy with strict operational constraints. From a deployment perspective, the system could be integrated into a centralized municipal cloud architecture, continuously ingesting streaming data from distributed sensor networks to forecast localized congestion events.
To be actionable, the system faces stringent latency requirements. If the model is predicting a transition into congestion three timesteps (e.g., 15 minutes) into the future, the end-to-end inference latency—including data ingestion, feature extraction, and model prediction—must be executed in near real-time (sub-second to low single-digit seconds). This operational reality underscores the value of our findings regarding tree-based models: while complex spatio-temporal GNNs offer marginal gains in specific contexts, highly optimized models like XGBoost provide exceptionally fast inference times that easily satisfy these low-latency constraints on standard hardware. Furthermore, even deep neural architectures can be systematically reduced and optimized (via techniques such as quantization or pruning) to guarantee real-time execution in production environments.

Finally, long-term deployment necessitates a rigorous retraining schedule and dynamic data integration. Traffic dynamics are not strictly stationary; they are subject to seasonality, evolving urban demographics, and long-term road construction, necessitating periodic offline retraining on a weekly or monthly cadence. More importantly, to adapt to acute, unforeseen topological changes, the models should be enriched with real-time external data streams. Utilizing platforms such as 511.org—which actively tracks real-time traffic incidents, lane closures, and accidents—would provide the models with critical contextual information to accurately determine congestion rates and sudden bottlenecks in the current network.

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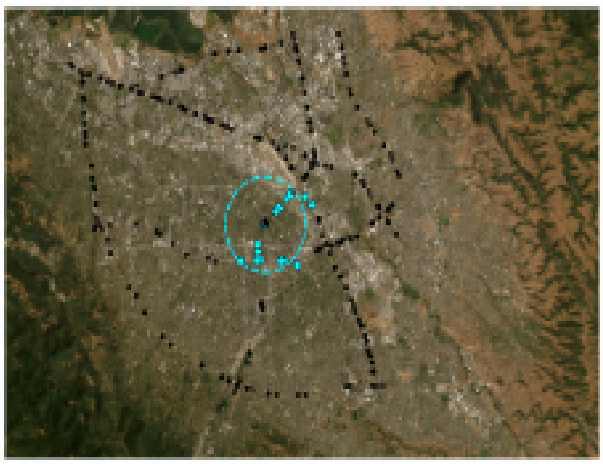
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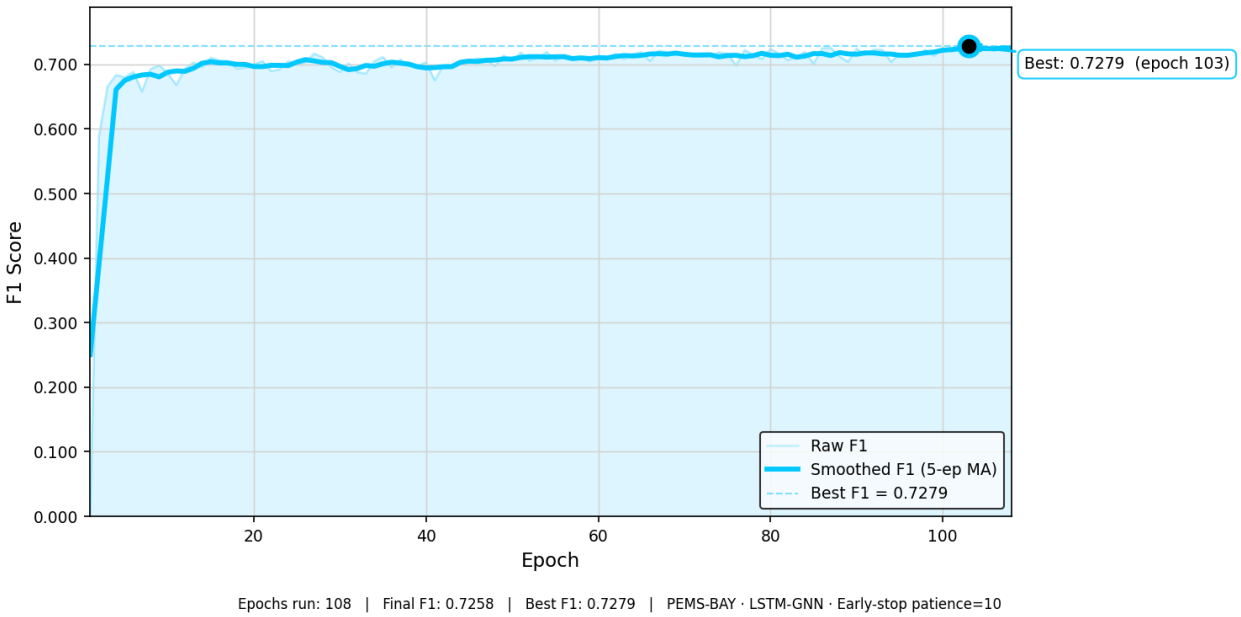
Figure 8a



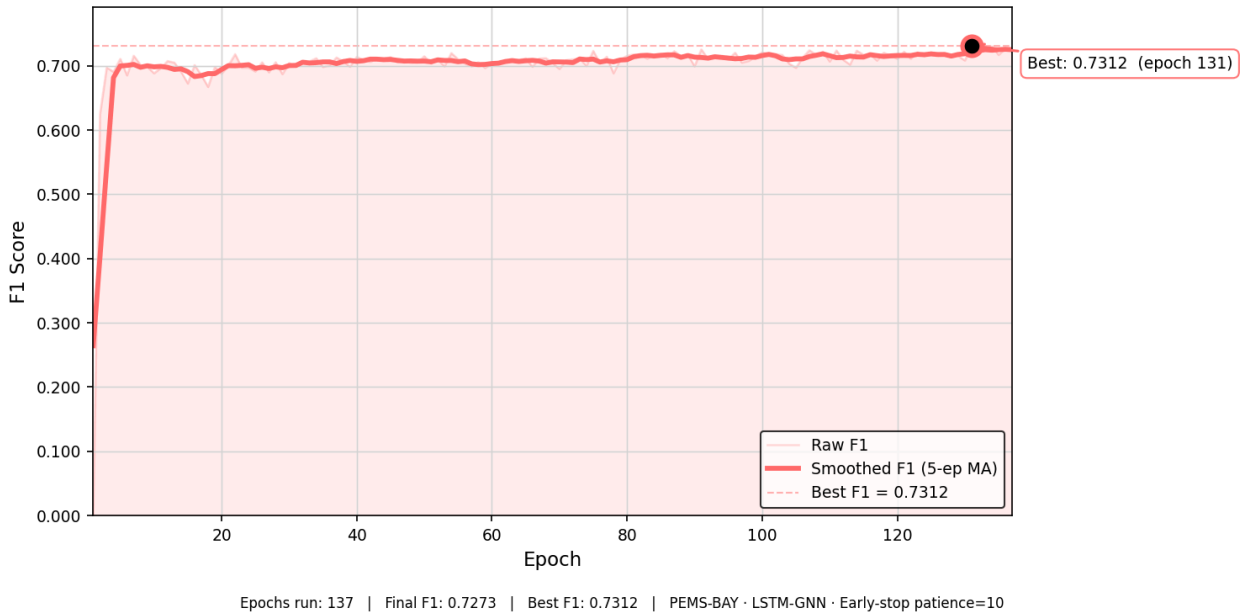
Figure 8b



Appendix A (Label each plot and add some explanation as to what is going on inside the plot)

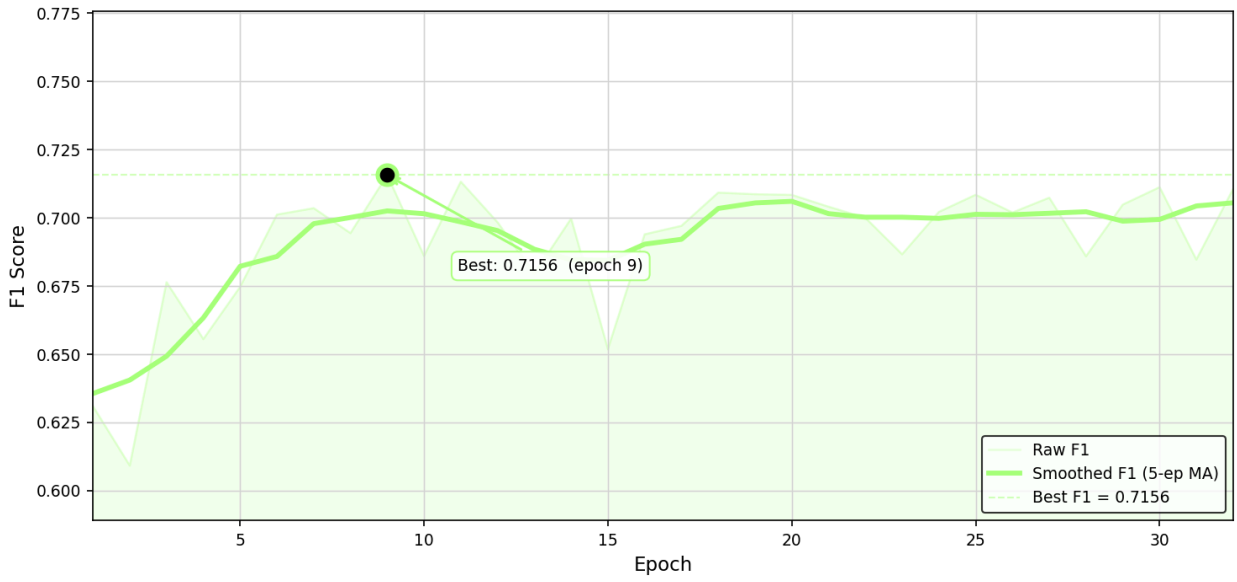


Plot of the GNN connecting only the points that are1 mile apart. F1 score ends up being similar to but not as high as the F1 score of the 2 mile radius graph. Despite starting to plateau quickly around the 4th epoch the graph still continues to slowly grow for a long amount of epochs.



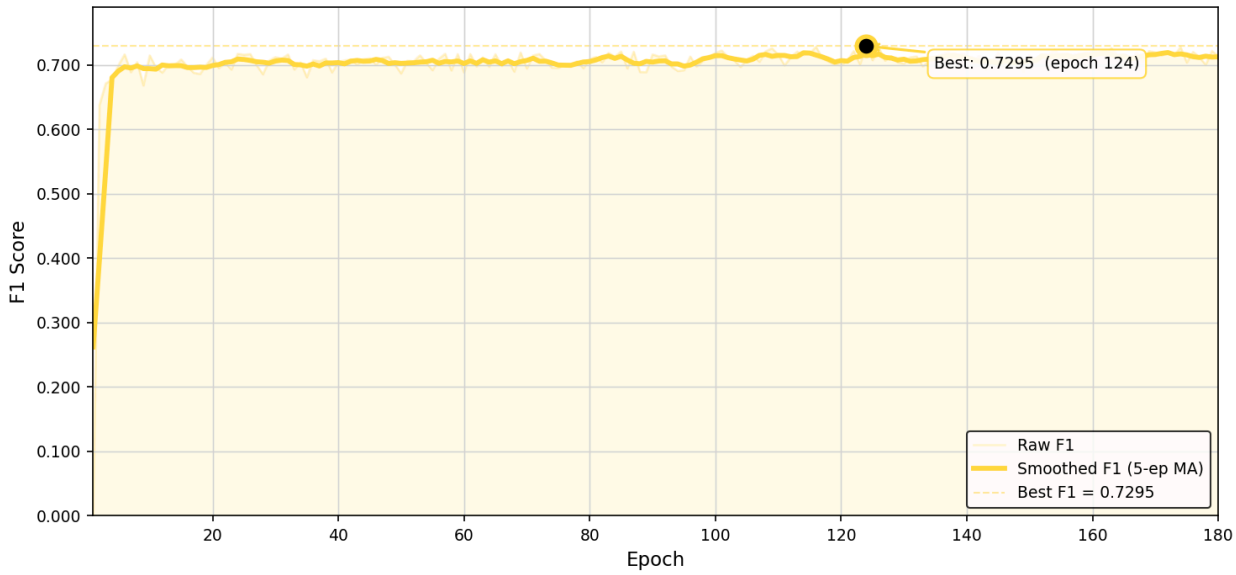
Epochs run: 137 | Final F1: 0.7273 | Best F1: 0.7312 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Plot of the GNN that connects nodes 2 miles apart. Does slightly better than the 1 mile radii. Shows growth similar to that of the 1 mile GNN but grows a little bit further. Overall still on an upwards trend suggesting potential growth.



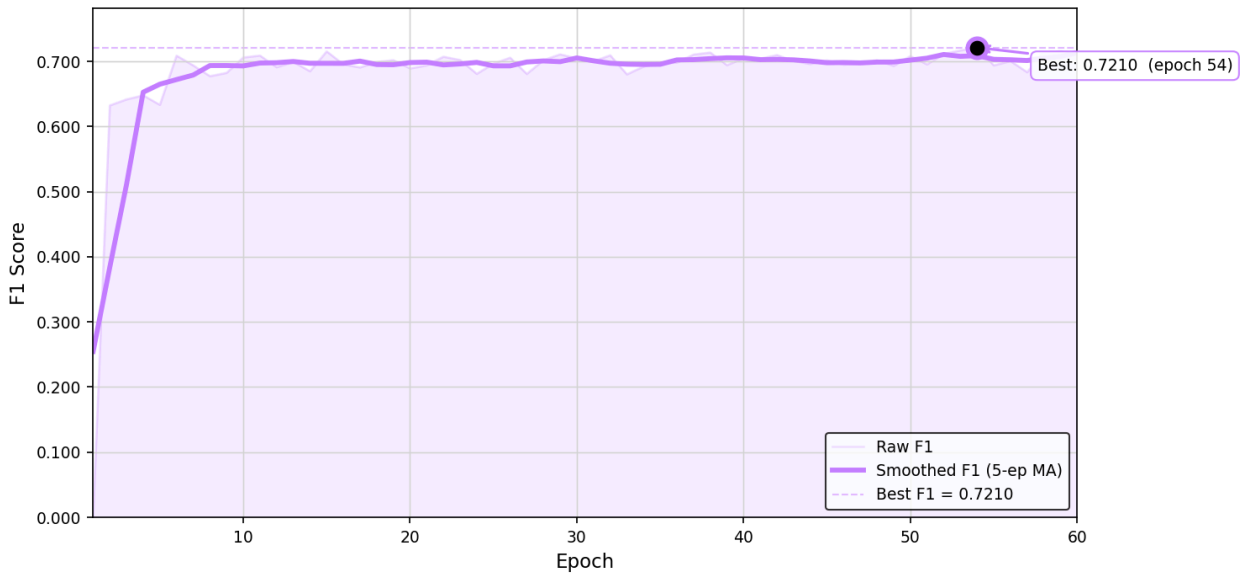
Epochs run: 32 | Final F1: 0.7105 | Best F1: 0.7156 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Graph of the 3 mile radius GNN. Ended much earlier than most other GNNs due to lack of growth during the training cycle.



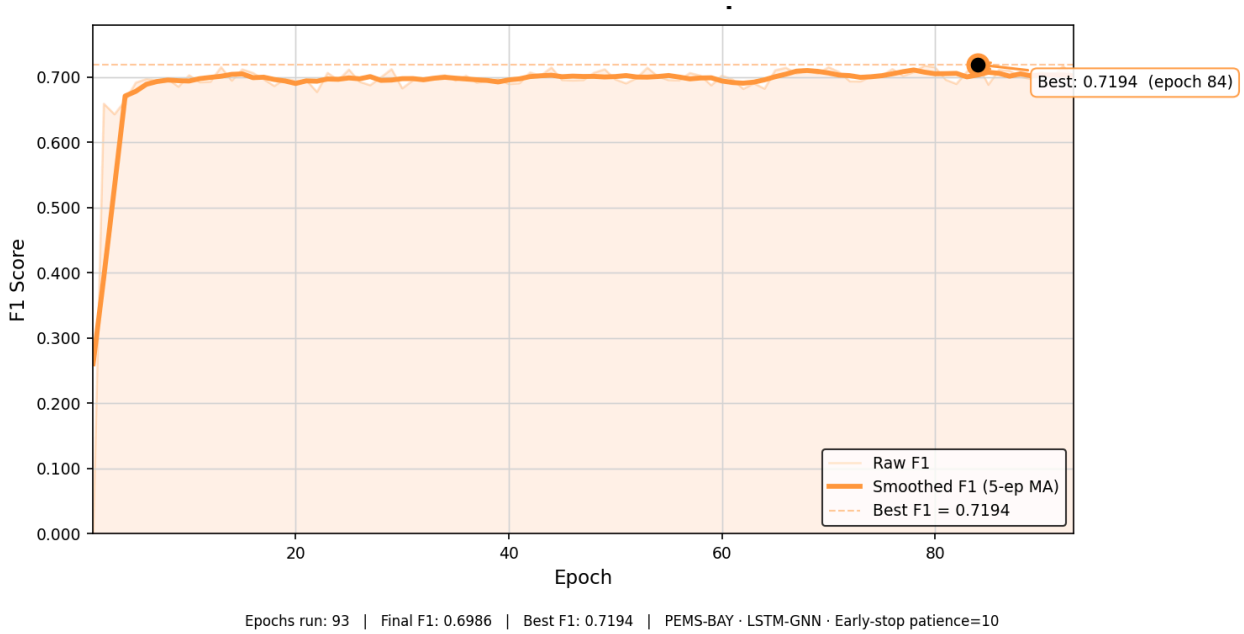
Epochs run: 180 | Final F1: 0.7137 | Best F1: 0.7295 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Graph of the 4 mile GNN. Trained for almost 200 epochs and performed similar to the 2 mile GNN. However the graph started to plateau much earlier and does not show signs of continued growth unlike the 2 mile GNN.



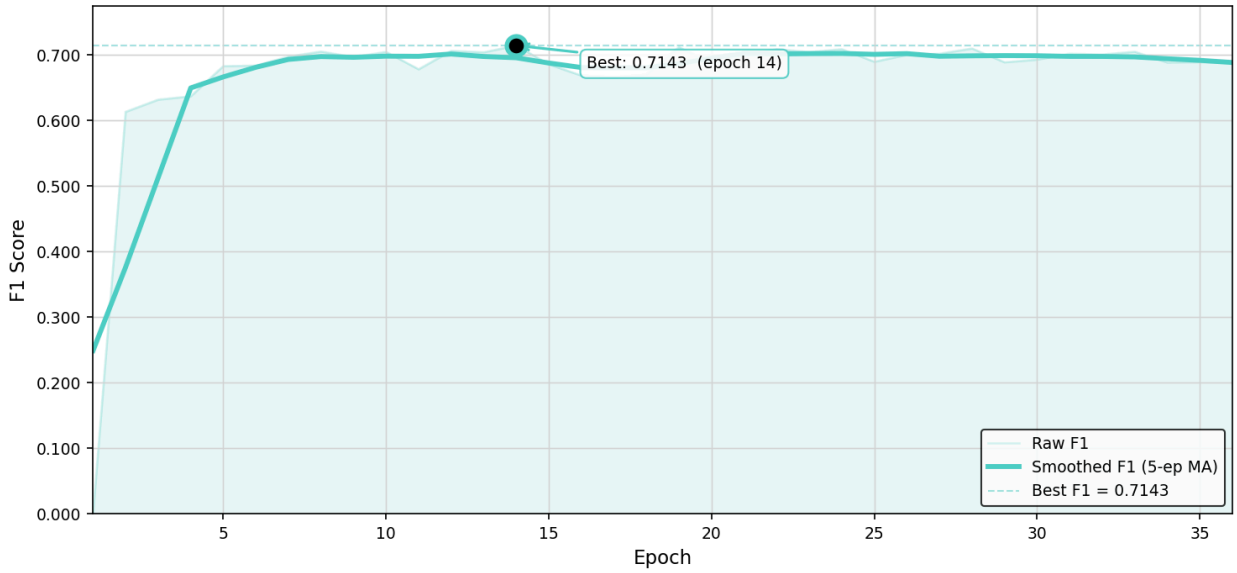
Epochs run: 60 | Final F1: 0.7121 | Best F1: 0.7210 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Graph of the 5 mile GNN. Similar to the 4 mile radius GNN however stops earlier and ends up performing slightly worse.



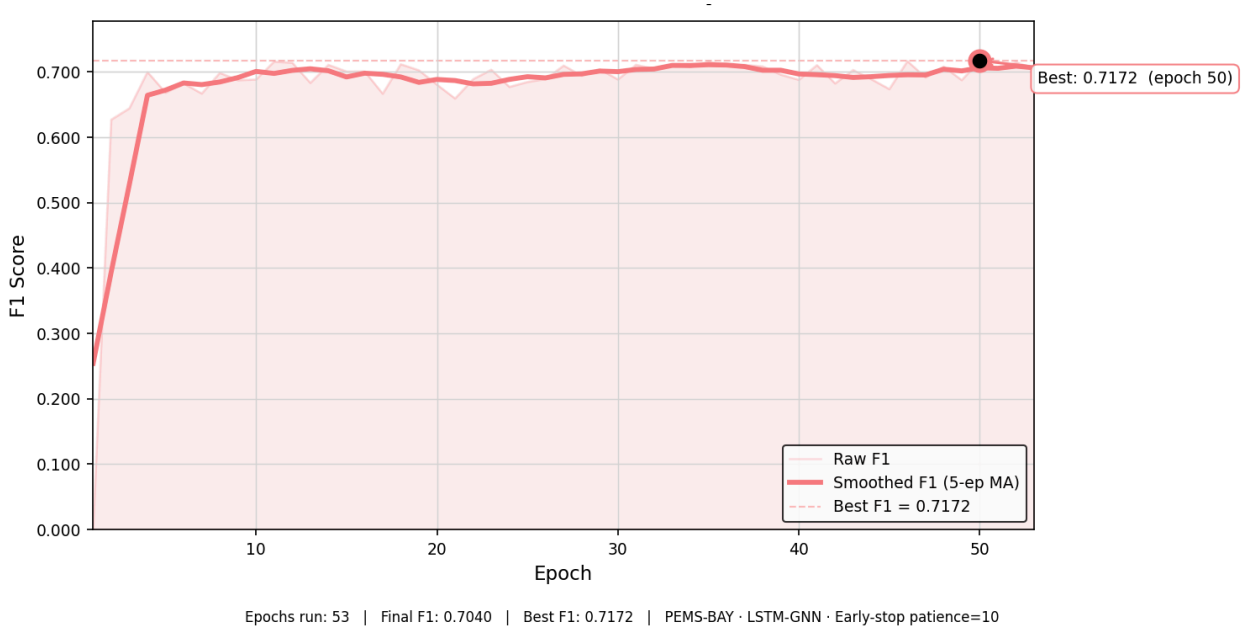
Epochs run: 93 | Final F1: 0.6986 | Best F1: 0.7194 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Graph of the 6 mile GNN. Continues the downward trend of the 5 mile GNN plateauing at a similar range.



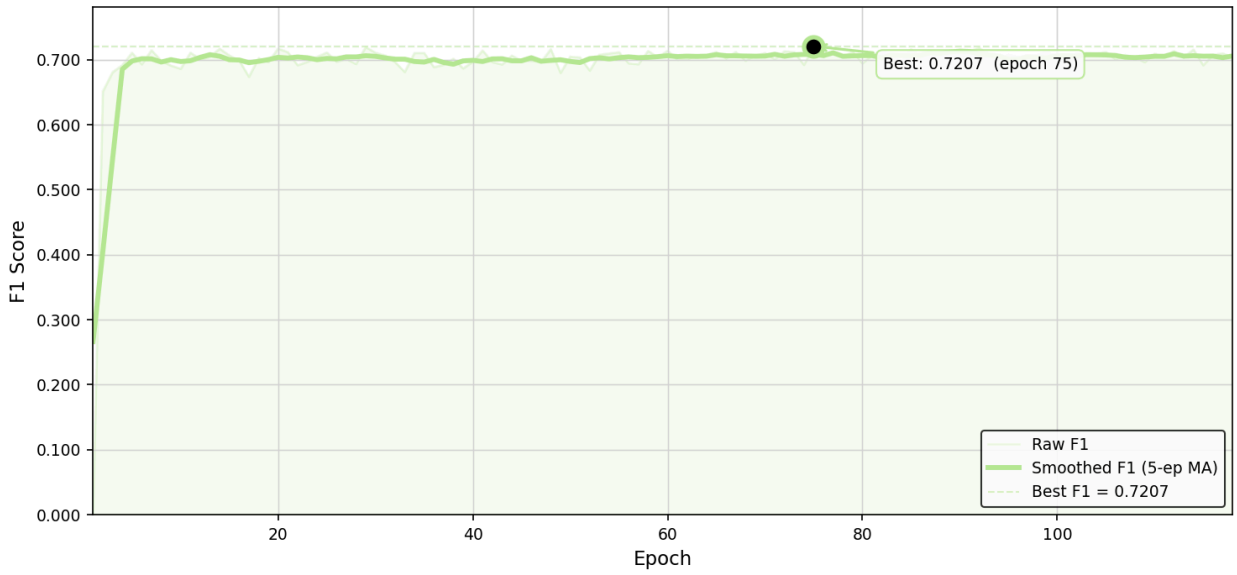
Epochs run: 36 | Final F1: 0.6882 | Best F1: 0.7143 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

The graph of the 7 mile GNN continues the trend of increasing miles decreasing the F1 score.



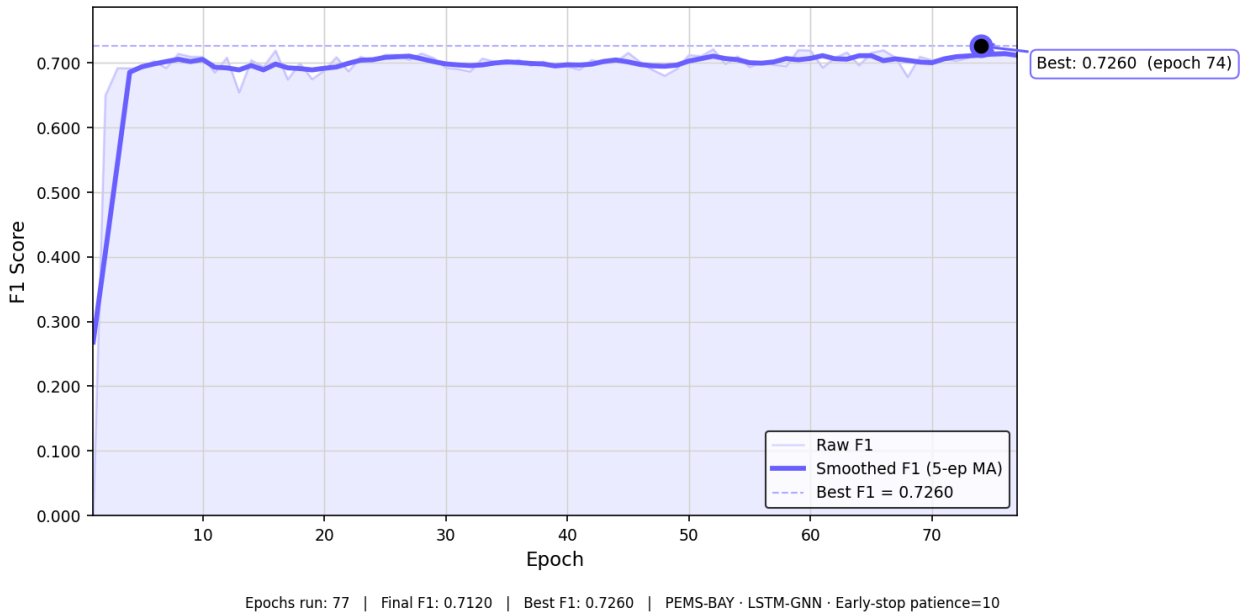
Epochs run: 53 | Final F1: 0.7040 | Best F1: 0.7172 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Graph of the 8 mile GNN counting with the trend of decreasing F1 scores despite performing the best with a lower epoch count. Suggesting that the increased noise from other nodes negatively impacts GNNs for longer epochs.



Epochs run: 118 | Final F1: 0.7037 | Best F1: 0.7207 | PEMS-BAY · LSTM-GNN · Early-stop patience=10

Graph of the 9 mile GNN. Plateaus similarly to the other GNNs but has a higher F1 score. Possibly due to larger GNNs having better F1 scores initially the 9 mile GNN performs better than smaller radii GNNs.



Graph of the 10 mile GNN. Similar to the 9 mile GNN however it has a slightly higher F1 score than the 9 mile GNN for potentially the same reason as mentioned in the 9 mile graph.